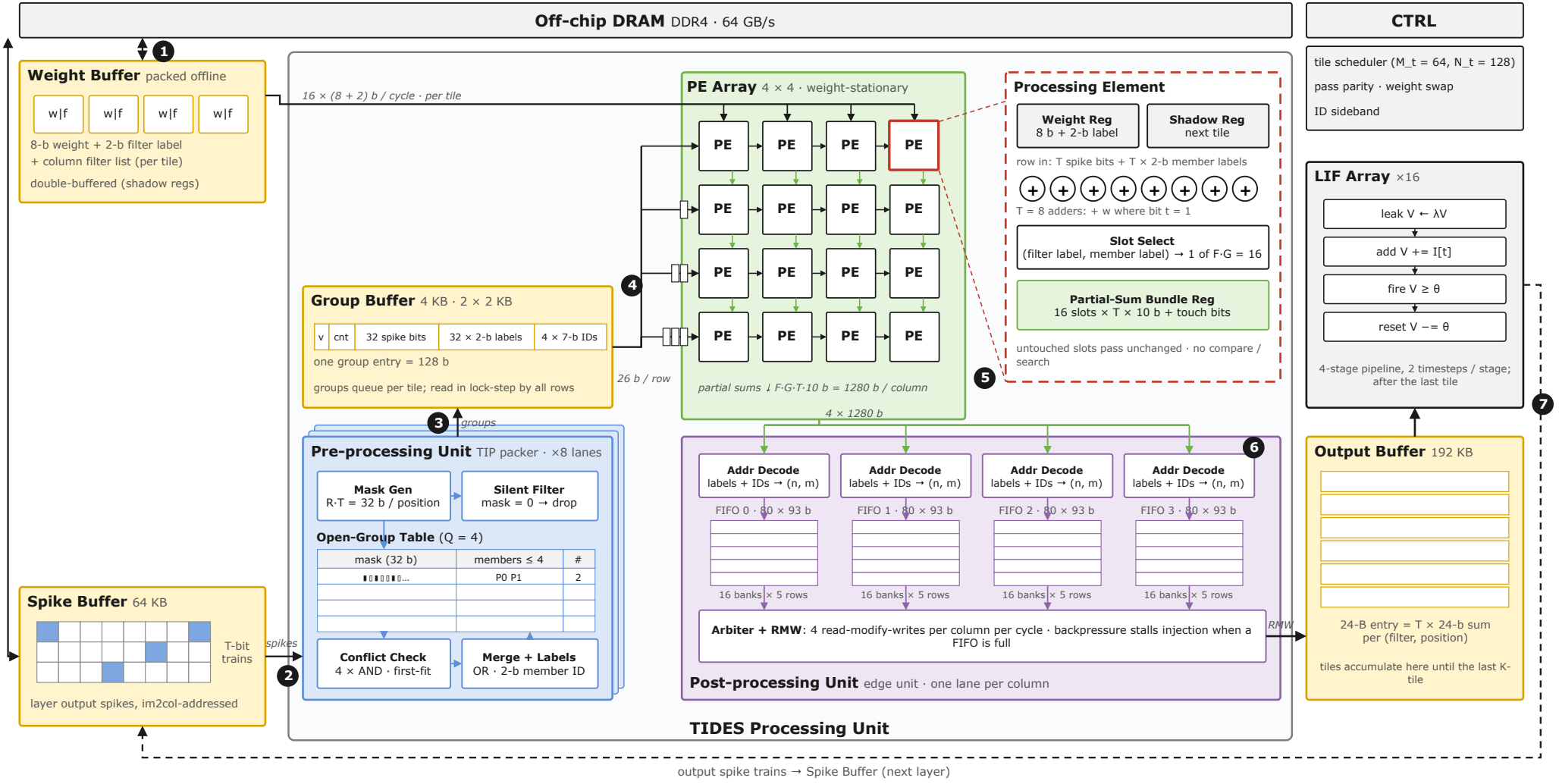

Off-chip DRAM DDR4 · 64 GB/s
CTRL
tile scheduler (M_t = 64, N_t = 128)
pass parity · weight swap
ID sideband
1
Weight Buffer packed offline
w|f
w|f
w|f
w|f
8-b weight + 2-b filter label
+ column filter list (per tile)
double-buffered (shadow regs)
16 × (8 + 2) b / cycle · per tile
PE Array 4 × 4 · weight-stationary
PE
Processing Element
Weight Reg
8 b + 2-b label
Shadow Reg
next tile
row in: T spike bits + T × 2-b member labels
T = 8 adders: + w where bit t = 1
Slot Select
(filter label, member label) → 1 of F·G = 16
Partial-Sum Bundle Reg
16 slots × T × 10 b + touch bits
untouched slots pass unchanged · no compare / search
LIF Array ×16
leak V ← λV
add V += I[t]
fire V ≥ θ
reset V −= θ
4-stage pipeline, 2 timesteps / stage; after the last tile
4
Group Buffer 4 KB · 2 × 2 KB
v
cnt
32 spike bits
32 × 2-b labels
4 × 7-b IDs
one group entry = 128 b
groups queue per tile; read in lock-step by all rows
26 b / row
partial sums ↓ F·G·T·10 b = 1280 b / column
4 × 1280 b
5
3
groups
Pre-processing Unit TIP packer · ×8 lanes
Mask Gen
R·T = 32 b / position
Silent Filter
mask = 0 → drop
Open-Group Table (Q = 4)
mask (32 b)
members ≤ 4
#
P0 P1
2
Conflict Check
4 × AND · first-fit
Merge + Labels
OR · 2-b member ID
6
Addr Decode
labels + IDs → (n, m)
FIFO 0 · 80 × 93 b
FIFO 1 · 80 × 93 b
FIFO 2 · 80 × 93 b
FIFO 3 · 80 × 93 b
16 banks × 5 rows
Arbiter + RMW: 4 read-modify-writes per column per cycle · backpressure stalls injection when a FIFO is full
RMW
Post-processing Unit edge unit · one lane per column
Output Buffer 192 KB
24-B entry = T × 24-b sum per (filter, position)
tiles accumulate here until the last K-tile
7
Spike Buffer 64 KB
T-bit trains
spikes
2
layer output spikes, im2col-addressed
TIDES Processing Unit
output spike trains → Spike Buffer (next layer)